

JIAJIE (JERRY) LUO

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Last update: May 21, 2026

EMPLOYMENT

Postdoctoral Scholar

October 2024 – Present

The Knowledge Lab

University of Chicago

Postdoctoral Mentor: Professor James Evans

Ph.D. Research Intern

June 2022 – September 2022

Mathematics, Statistics, and Data Science

Pacific Northwest National Laboratory

Mentors: Dr. Tegan Emerson; Dr. Gregory Henselman-Petrusek Roek

EDUCATION

University of California, Los Angeles

September 2019 – June 2024

Ph.D. in Mathematics.

Thesis Title: Topics in Persistent Homology and Complex Social Systems

Advisor: Professor Mason Porter

University of California, Santa Barbara

September 2017 – June 2019

M.A. in Mathematics.

Thesis Title: On Abstract Witt Rings and Quadratic Extensions

Advisor: Professor Bill Jacob

University of California, Santa Barbara

September 2014 – June 2017

College of Creative Studies

B.S. in Mathematics, *Highest Honors*

Faculty Advisor: Professor Jeffrey Stopple

RESEARCH INTERESTS

Topological Data Analysis, Persistent Homology and Applications, Network Science, Complex Systems, Opinion Dynamics Modelling

PREPRINTS & PUBLICATIONS

1. L. Thompsky, Y. Wu, M. A. Porter, **J. Luo**, *A Bounded-Confidence Model of Opinion Dynamics with Adaptive Interaction Probabilities*, arXiv:2605.20418
2. **J. Luo**, G. Henselman-Petrusek, *Interval Decomposition of Infinite Persistence Modules over a Principal Ideal Domain*, To Appear in Proceedings of the American Mathematical Society, arXiv:2511.07614
3. G. J. Li, **J. Luo**, W. Chu, *Bounded-Confidence Models of Multi-Dimensional Opinions with Topic-Weighted Discordance*, Published in *SIAM Journal on Applied Dynamical Systems*.

4. **J. Luo**, G. Henselman-Petrusek, *Interval Decomposition for Persistence Modules Over a Principal Ideal Domain*, Published in *Foundations of Computational Mathematics*.
5. G. J. Li*, **J. Luo***, M. A. Porter (*Equal Contribution), *Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds*, Published in *SIAM Journal on Applied Dynamical Systems*.
6. A. Hickok*, B. Jarman*, M. C. Johnson*, **J. Luo***, M. A. Porter (*Equal Contribution), *Persistent Homology for Resource Coverage: A Case Study of Access to Polling Sites*, Published in *SIAM Review*.
7. V. Chayes, K. Miller, R. Bhalerao, **J. Luo**, W. Zhu, A. Bertozzi, W. Liao, S. Osher, *Pre-Processing and Classification of Hyperspectral Imagery Via Selective inpainting*, Published in *ICASSP2017*.

EXPOSITORY ARTICLES

G. J. Li, **J. Luo**, K. Peng, and M. A. Porter. *Using Mathematics to Study How People Influence Each Other's Opinions*, Published in *Frontiers for Young Minds*.

AWARDS, HONORS & FELLOWSHIPS

Pacific Journal of Mathematics Dissertation Prize	2024
ModELing and uNdersTanding human behaviOR (MENTOR) Fellowship	2021–2022
College of Creative Studies Commencement Speaker	2017
Adil Yaqub is my Hero Scholarship	2016

TALKS & PRESENTATION

SIAM Annual Meeting (AN26)	July 2026
Advances in Modeling and Inference for Dynamics on Networks (Minisymposium)	
Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Invited Talk)	
LANS Seminar, Argonne National Laboratory	April 2026
Generalizations and Extensions of Bounded-Confidence Models (Invited Talk)	
IMSI Workshop: The Geometric Realization of AATRn	August 2025
Interval Decomposition of Persistence Modules over a Principal Ideal Domain (Poster + Lightning Talk)	
IMSI Workshop: Emergent Behavior in Complex Systems of Interacting Agents	March 2025
Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Poster Session)	
Southern California Applied Mathematics Symposium (SOCAMS)	April 2024
Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Contributed Talk)	
Graduate Student Topology and Geometry Conference (GSTGC2024)	April 2024
Interval Decomposition of Persistence Modules over a Principal Ideal Domain (Poster Session)	
Joint Mathematics Meetings (JMM 2024)	January 2024
AMS Special Session on Complex Social Systems I	
Persistent Homology for Assessing Facility Placement (Invited Talk)	
2023 Algorithms for Threat Detection PI Workshop (ATD2023)	October 2023
Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Contributed Talk)	
Computation Persistence Workshop (ComPer23)	September 2023
Interval Decomposition for Persistence Modules of Free Abelian Groups (Contributed Talk)	
SIAM Conference on Applications of Dynamical Systems (DS23)	May 2023
Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Contributed Talk)	

Southern California Applied Mathematics Symposium (SOCAMS)	April 2023
Persistent Homology for Resource Coverage: A Case Study of Access to Polling Sites (Contributed Talk)	
SIAM Workshop on Network Science (NS22)	September 2022
Bounded-Confidence Models with Adaptive Confidence Bounds (Contributed Talk)	
Virtual Research Symposium, Pacific Northwest National Laboratory.	August 2022
Topological Data Analysis and Machine Learning (Contributed Talk)	

TEACHING EXPERIENCE

As Graduate Student Instructor (UCLA)

Math 110A: Abstract Algebra	Winter 2024
Math 115A: Linear Algebra (proof-based)	Winter 2023

As Graduate Teaching Assistant (UCLA)

Math 31AL: Differential and Integral Calculus Laboratory	Winter 2021
Math 115A: Linear Algebra (proof-based)	Fall 2020, Spring 2021
Math 31B: Integration and Infinite Series	Spring 2020
Math 33A: Linear Algebra and Application	Winter 2020, Fall 2020, Spring 2021
Math 3B: Calculus for Life Sciences II	Fall 2019, Winter 2021

As Graduate Teaching Assistant (UCSB)

Math 117: Methods of Analysis	Spring 2019
Math 108A: Introduction to Linear algebra (proof-based)	Winter 2019
Math 4A: Linear Algebra and Applications	Fall 2018
MATH 100B: Mathematics for Elementary Teaching II	Summer 2018
Math 34B: Calculus for Social Sciences II	Winter 2018, Spring 2018
Math 34A: Calculus for Social Sciences I	Fall 2017

STUDENT MENTORING

Research Mentoring (Students from UChicago):

Ziqian Zhao — A Bounded-Confidence Model with Network Rewiring	Fall 2025 – Present
Joy Yang — The Effects of Topic Shifts on Researchers	Fall 2025 – Present
Xiang (Sherry) Ding — The Effects of Topic Shifts on Researchers	Fall 2025 – Present
William Flowers — Bounded-Confidence Models of Opinion Dynamics	Fall 2024 – Present

Research Mentoring (Students from UCLA):

Yuxuan Wu — A Bounded-Confidence Model with Adaptive Edge Weights	Summer 2024 – Spring 2026
Leila Thompsky — A Bounded-Confidence Model with Adaptive Edge Weights	Fall 2023 – Spring 2026
Amos Ancell — Persistent Homology for Resource Coverage	Fall 2023 – Spring 2024
Ruyi Lu — Bounded-Confidence Models on Random Configuration Models	Winter 2023 – Fall 2023
Xinyue (Serena) Li — Persistent Homology for Resource Coverage	Winter 2023 – Spring 2023
Xiaohe (Haley) Zhang — Bounded-Confidence Models with Repulsion	Winter 2022 – Spring 2022

Directed Reading Program:

DRP Committee	Fall 2021 – Spring 2024
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Students:

Yuxuan (Yolanda) Wu — Models of Opinion Dynamics	Spring 2024
Leila Thompsky — Complex Social Systems	Fall 2023
Amos Ancell — Applied Topology, Persistent Homology	Winter 2023 – Spring 2023
Xinyue (Serena) Li — Applied Topology, Persistent Homology	Fall 2022 – Winter 2023
Ruyi Lu — Opinion Dynamics on Networks	Fall 2022 – Winter 2023
Haoyang Lyu — Applied Topology, Persistent Homology	Winter 2022 – Spring 2022
Chenxin (Amy) Shen — Applied Topology, Persistent Homology	Fall 2021 – Spring 2022
Xiaohe (Haley) Zhang — Opinion Dynamics on Networks	Fall 2021 – Winter 2022

Tanishq Bhatia — Topics in Persistent Homology

Winter 2021 – Spring 2021

Other Mentoring:

Mentor for UCLA Applied Mathematics REU (ATD Traffic Challenge)

Summer 2021

Students:

- Matthew Hudes (Tufts University)
- Naji Sarsam (UCLA)
- Chenxin (Amy) Shen (UCLA)
- Wenwen Tang (USC)

MISCELLANEOUS

Citizenship: United States

Programming Experience: Python, MATLAB, R., C++

Languages: Chinese (Mandarin), English.